Starter

1. Find $\frac{\mathrm{d}}{\mathrm{d}x}(xy)$
2. Find $\frac{\mathrm{d}}{\mathrm{d}x}(y \sin x)$
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4. Find $\frac{\mathrm{d}}{\mathrm{d}x}(\sinh(x)e^y)$
5. Find $\frac{\mathrm{d}}{\mathrm{d}x}(\ln(x)\tan(2y))$

Starter

1. Find $\frac{\mathrm{d}}{\mathrm{d}x}(xy) = x\frac{\mathrm{d}y}{\mathrm{d}x} + y$
2. Find $\frac{\mathrm{d}}{\mathrm{d}x}(y\sin x) = y\cos x + \sin x\frac{\mathrm{d}y}{\mathrm{d}x}$
3. Find $\frac{\mathrm{d}}{\mathrm{d}x}(xy^2) = y^2 + 2xy\frac{\mathrm{d}y}{\mathrm{d}x}$
4. Find $\frac{\mathrm{d}}{\mathrm{d}x}(\sinh(x)e^y) = \cosh(x)e^y + \sinh(x)e^y\frac{\mathrm{d}y}{\mathrm{d}x}$
5. Find $\frac{\mathrm{d}}{\mathrm{d}x}(\ln(x)\tan(2y)) = \frac{\tan(2y)}{x} + 2\ln(x)\sec^2(2y)\frac{\mathrm{d}y}{\mathrm{d}x}$

What is a differential equation?

Technically, this is an equation involving a variable, x, a function of x (usually 'y'), and the derivative of the function $\frac{\mathrm{d}y}{\mathrm{d}x}$.

The goal is to work out what function satisfies that equation.

It's often easiest to imagine these as a **vector field**.

EG.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x$$

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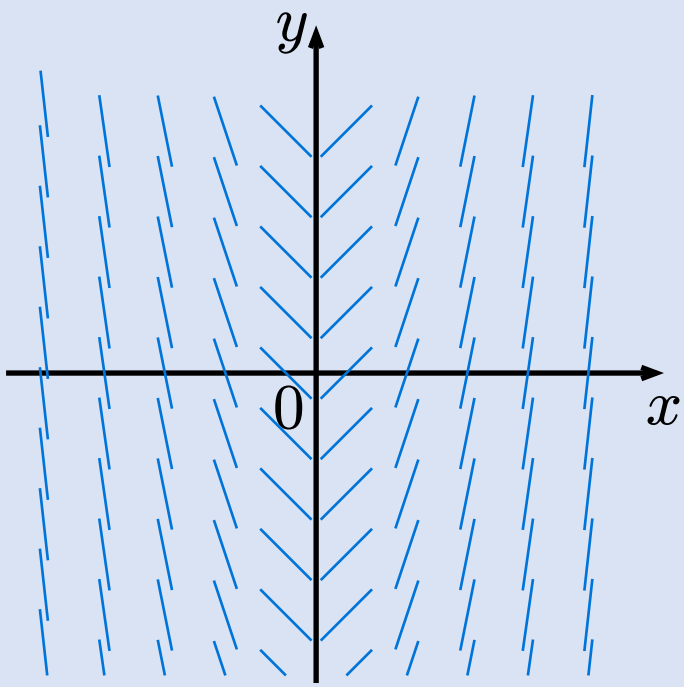
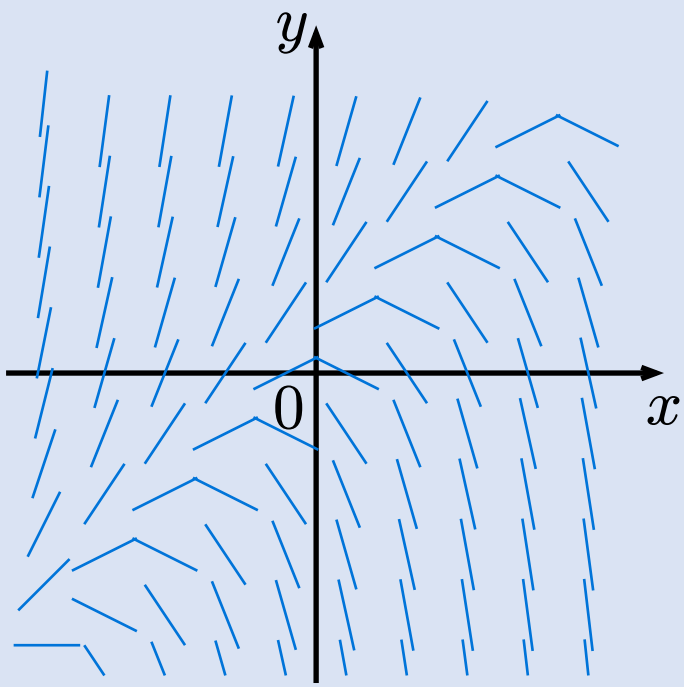


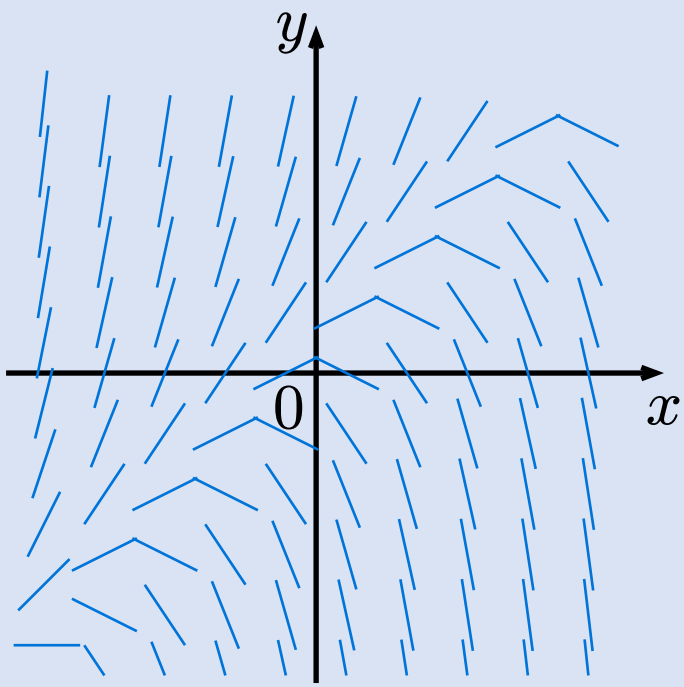
The gradient at any point on this graph can be calculated as $2x$, so we'll see a quadratic if we choose any specific point.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y - x$$



Every point within this space has a direction and a magnitude.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y - x$$



Every point within this space has a direction and a magnitude.

If we want to find a particular function in here we can imagine dropping a point into this field and watching where it goes.

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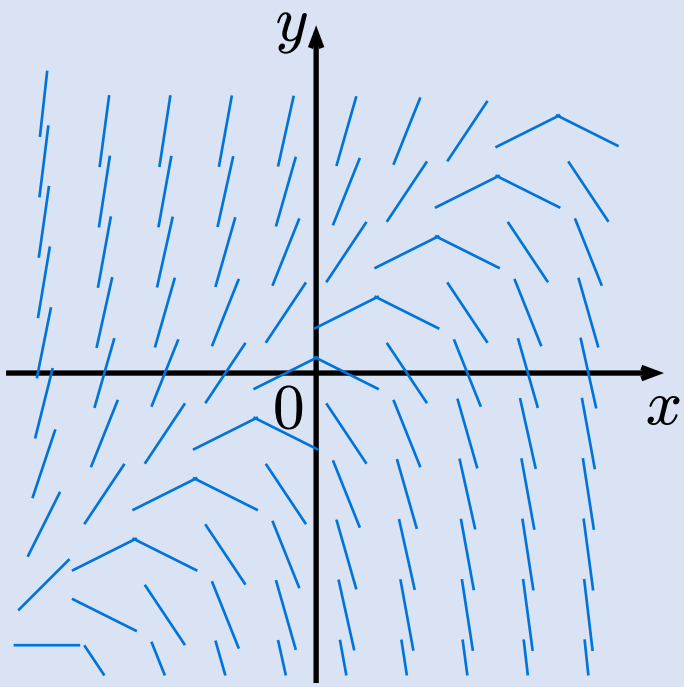

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If we want to find a particular function in here we can imagine dropping a point into this field and watching where it goes.

More formally, this is **integration**.
What does the $+c$ represent then?

Differiental Equations

What kinds of shapes do these more complicated equations make?

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2xy$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x + y$$

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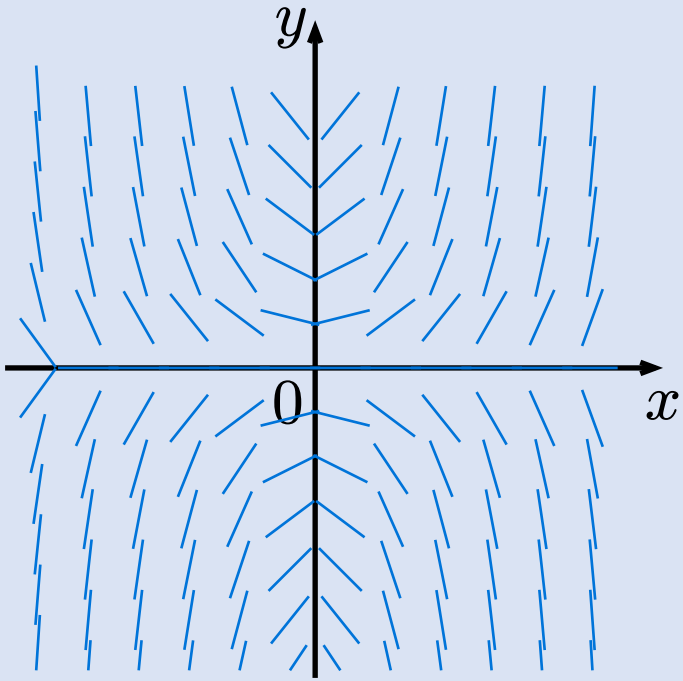

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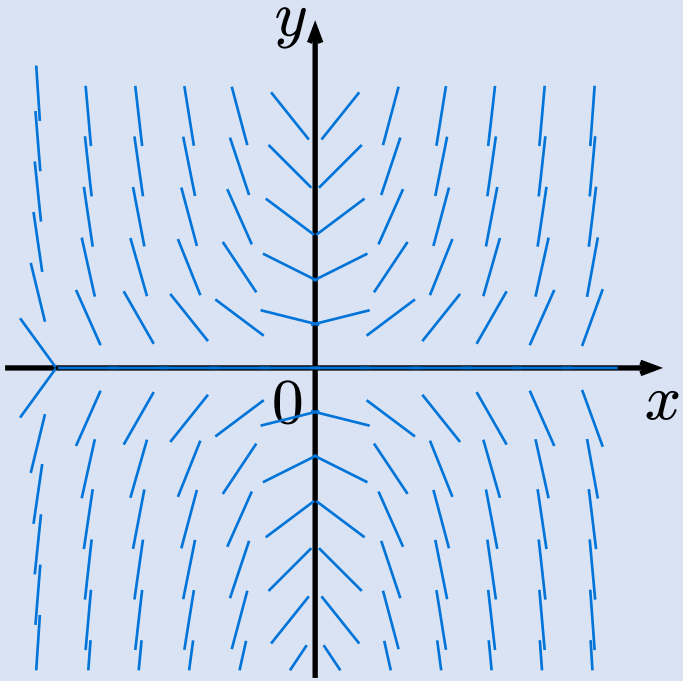

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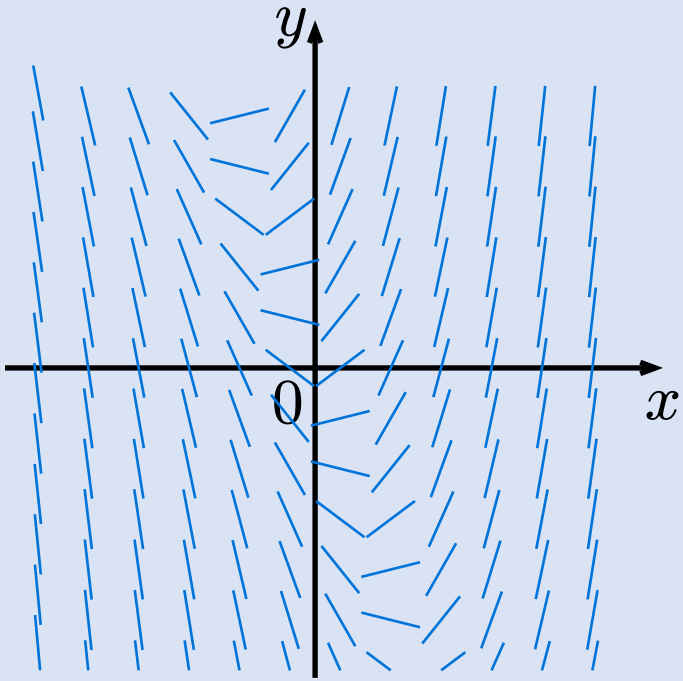

Differential Equations

Sometimes they are more complicated.

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \frac{\mathrm{d}y}{\mathrm{d}x} + y = e^x$$

This means that there is some function y such that the sum of y, the first derivative and the second derivative makes e^x

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We'll look at techniques to solve a small number of different types of DEs on this course, but they do go much further.

So far in A Level Maths, you've used **Separation of Variables**

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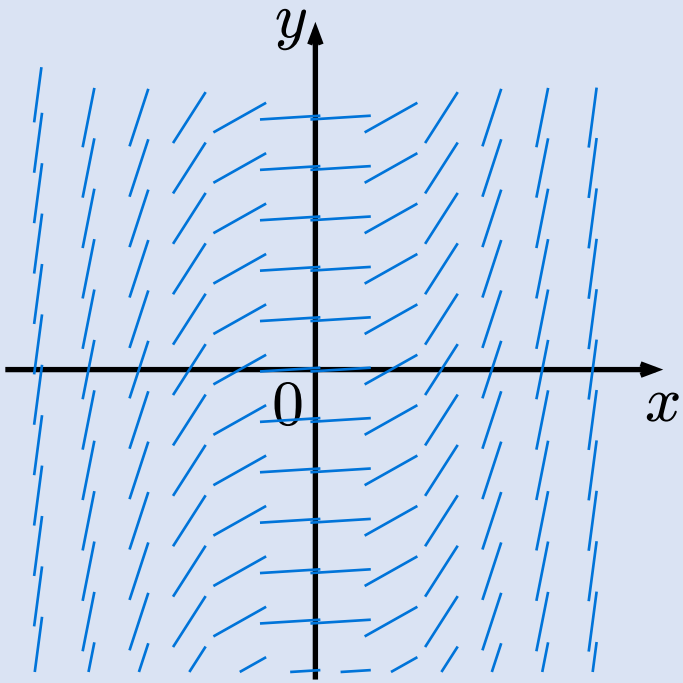

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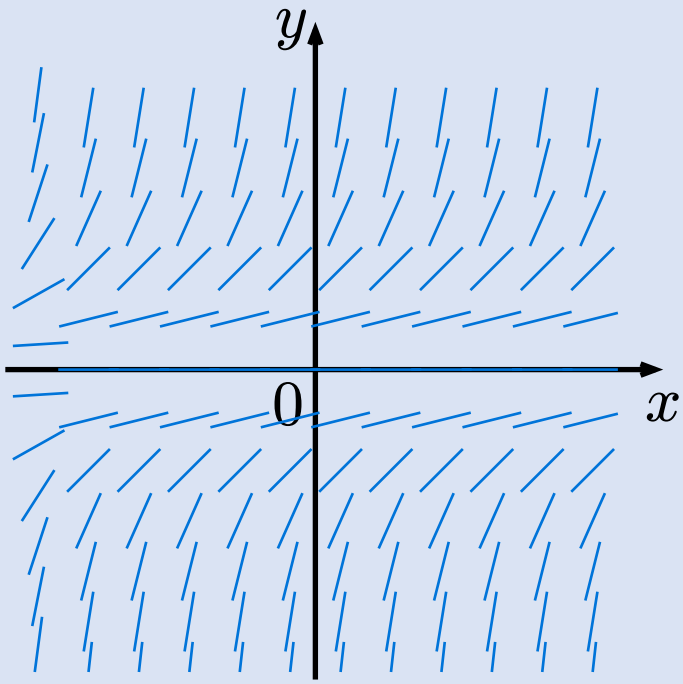

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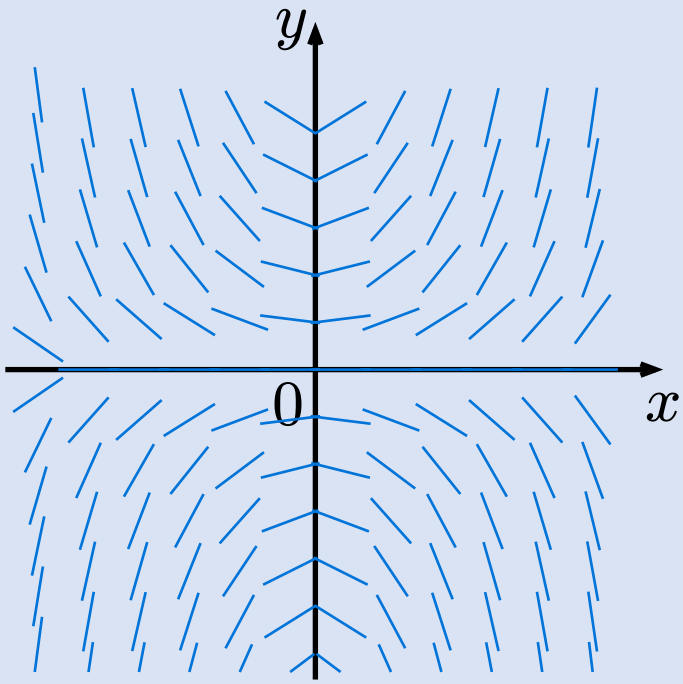

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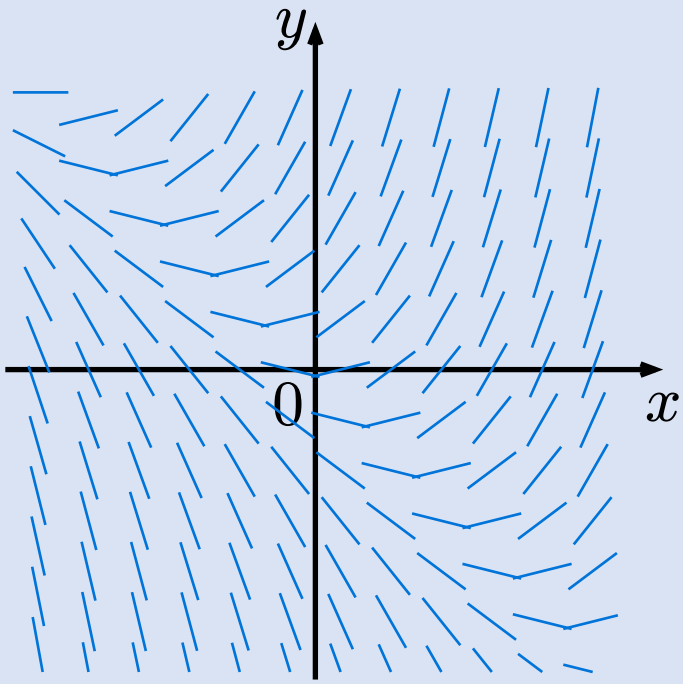

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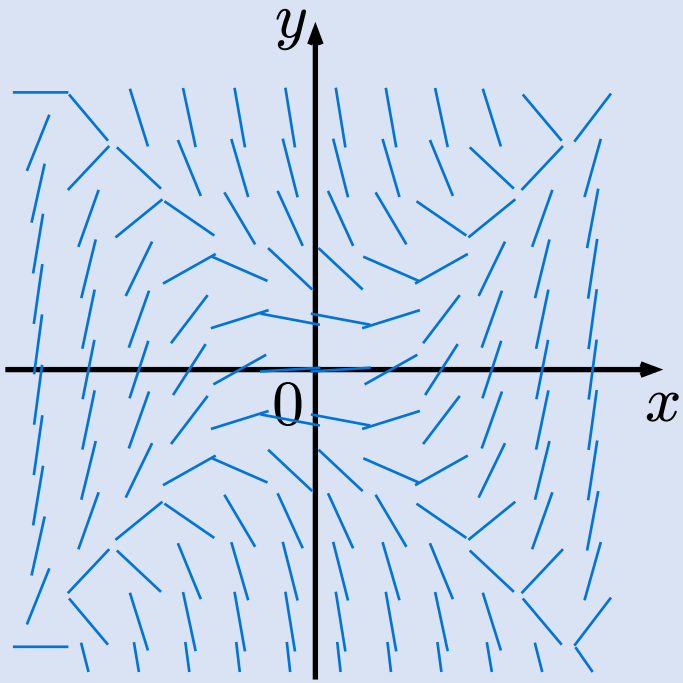

Separation of Variables - Recap

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$$y\frac{\mathrm{d}y}{\mathrm{d}x} = 2x \Rightarrow \qquad \int y \ \mathrm{d}y = \int 2x \ \mathrm{d}x$$

$$\therefore \frac{1}{2}y^2 = x^2 + c \Rightarrow x^2 - \frac{1}{2}y^2 + c = 0$$

Given that $y = 0$ when $x = 1$, find c.

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Given that $y = 0$ when $x = 1$, find c.

$$1 - 0 + c = 0 \Rightarrow c = -1 \Rightarrow x^2 - \frac{1}{2}y^2 - 1 = 0$$

Recap: You Try

Find the solution to each differential equation, given the initial conditions.

1. $\frac{\mathrm{d}y}{\mathrm{d}x} = xy, x = 0, y = 2$
2. $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{e^y}{3}, x = e, y = -1$
3. $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\sin x}{y}, x = \frac{\pi}{2}, y = 1$
4. $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1+y^2}{2}, x = 1, y = 0$
5. $\frac{\mathrm{d}y}{\mathrm{d}x} = y - x, x = 0, y = 3$

solutions

$\frac{\mathrm{d}y}{\mathrm{d}x} = xy, x = 0, y = 2, x = e, y = -1$

$$\int \frac{1}{y}\,\mathrm{d}y = \int x\,\mathrm{d}x$$

$$\ln y = \frac{1}{2}x^2 + c$$

$$y = e^{\frac{1}{2}x^2 + c} \equiv Ae^{\frac{1}{2}x^2}$$

$$y|_{(0,2)} = 2 = Ae^0 \Rightarrow A = 2$$

$$\therefore y = 2e^{\frac{1}{2}x^2}$$

solutions

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{e^y}{3}$

$$\int e^{-y}\,\mathrm{d}y = \int \frac{1}{3}\,\mathrm{d}x$$

$$-e^{-y} = \frac{1}{3}x + c \Rightarrow y = -\ln\left|c - \frac{1}{3}x\right|$$

$$y\big|_{(e,-1)} = -1 = -\ln\left(c - \frac{1}{3}e\right)$$

$$e = c - \frac{1}{3}e \Rightarrow c = \frac{4}{3}e$$

$$\therefore y = -\ln\left(\frac{4}{3}e - \frac{1}{3}x\right)$$

solutions

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\sin x}{y}, x = \frac{\pi}{2}, y = 1$

$$\int y \ \mathrm{d}y = \int \sin x \ \mathrm{d}x \Rightarrow \frac{1}{2}y^2 = -\cos x + c$$

$$y^2 = c - 2\cos x \ |_{(\frac{\pi}{2},1)} = 1 = c - 2 \times 0 \Rightarrow c = 1$$

$$\therefore y^2 = 1 - 2\cos x \ \text{ or } \ y = \sqrt{1 - 2\cos x}$$

solutions

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1+y^2}{2}, x = 1, y = 0$

$$\int \frac{1}{1+y^2}\mathrm{d}y = \int \frac{1}{2}\,\mathrm{d}x \Rightarrow \tan y = \frac{1}{2}x + c$$

$$@(1,0) = 0 = \frac{1}{2} + c \Rightarrow c = -\frac{1}{2}$$

$$\therefore \tan y = \frac{1}{2}x - \frac{1}{2} \text{ or } y = \arctan\left(\frac{1}{2}x - \frac{1}{2}\right)$$

solutions

$\frac{\mathrm{d}y}{\mathrm{d}x} = y - x, x = 0, y = 3$

Can't be solved right now. We can't separate the variables.

Sorry!

We'll look at some options for these now and next lesson.

From the Starter

Find $\frac{\mathrm{d}}{\mathrm{d}x}(xy)$

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$$\therefore \int x\frac{\mathrm{d}y}{\mathrm{d}x} + y \ \mathrm{d}x = xy + c$$

So if we had:

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + y = x^2$$

We could integrate both sides, and get:

$$\int x\frac{\mathrm{d}y}{\mathrm{d}x} + y\mathrm{d}x = \int x^2 \Rightarrow xy = \frac{1}{3}x^3 + c \Rightarrow y = \frac{1}{3}x^3 + \frac{c}{x}$$

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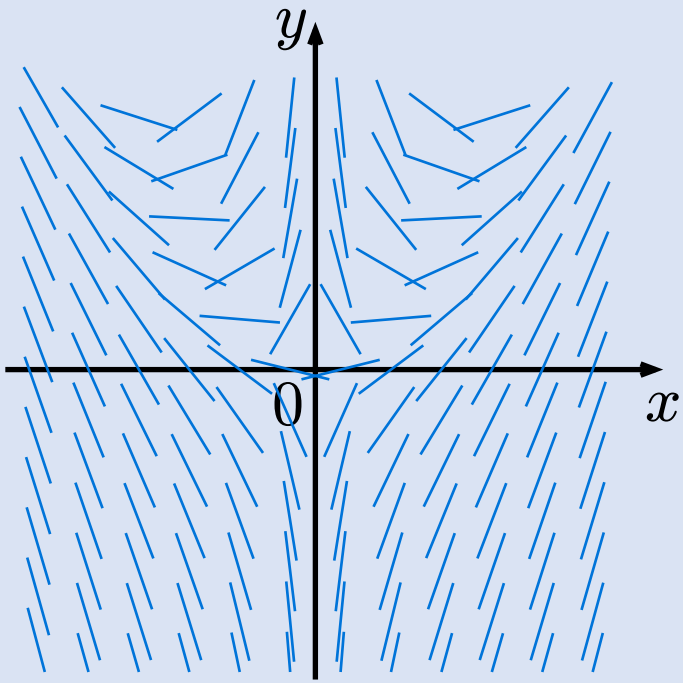
$$x\frac{\mathrm{d}y}{\mathrm{d}x} + y = x^2$$



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$$\therefore \int \frac{\mathrm{d}y}{\mathrm{d}x}\sin x + y\cos y \ \mathrm{d}x = y\sin x + c$$

Which is effectively reversing the product rule!

Example

Find the General Solution to:

$$2xy\frac{\mathrm{d}y}{\mathrm{d}x} + y^2 = x^2$$

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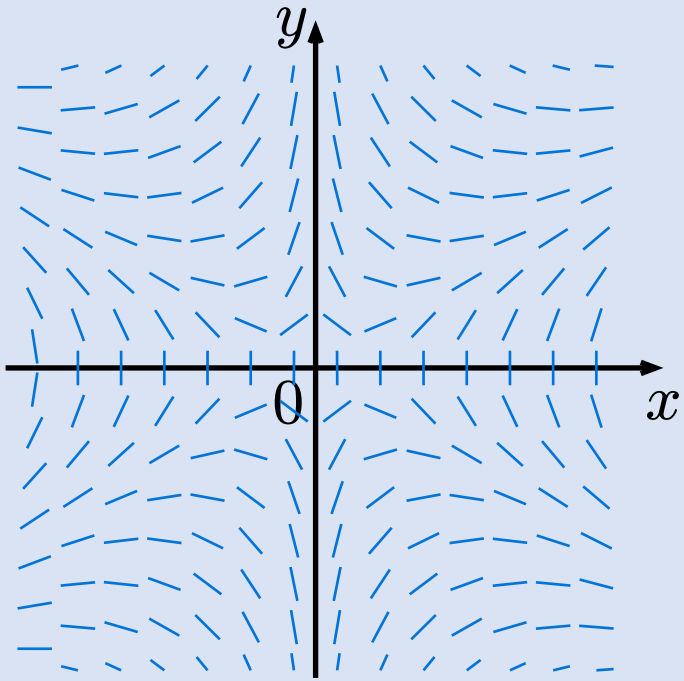
$$y^2 \to 2y\frac{\mathrm{d}y}{\mathrm{d}x}, x \to 1, \therefore \text{ product rule!}$$

$$\int 2xy\frac{\mathrm{d}y}{\mathrm{d}x} + y^2 \ \mathrm{d}x = \int x^2 \ \mathrm{d}x$$

$$xy^2 = \frac{1}{3}x^3 + c \Rightarrow y^2 = \frac{1}{3}x^2 + \frac{c}{x}$$



Practice Time!

Find the general solutions by reversing the product rule:

1. $x^3 \frac{\mathrm{d}y}{\mathrm{d}x} + 3x^2 y = \frac{1}{x}$
2. $\sqrt{x}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{2\sqrt{x}} y = x + 1$
3. $\cosh(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \sinh(x) y = 2$
4. $\ln(x) e^y + \frac{e^y}{x} = x$
5. $x^3 y^2 \frac{\mathrm{d}y}{\mathrm{d}x} + x^2 y^3 = 2$
6. $x^2 \frac{\mathrm{d}y}{\mathrm{d}x} + 4xy = 3$

Alternative Method

Use the substitution $x = y^2$ to find the general solution to

$$2xy\frac{\mathrm{d}y}{\mathrm{d}x} + y^2 = 2x + 1$$